

EE3621: Digital Signal Processing

Lecture 3: The Discrete-Time Fourier Transform (DTFT) (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Motivation: Time-domain limitations, continuous spectrum, and periodicity.
- **05:00 – 12:00 (7 mins):** DTFT & IDTFT Mathematical Definitions, Convergence Criteria.
- **12:00 – 22:00 (10 mins):** DTFT of Common Signals: Rectangular Pulse (Dirichlet Kernel), Exponentials, Impulse, and DC.
- **22:00 – 35:00 (13 mins):** Key Properties of the DTFT (Linearity, Shifting, Modulation, Symmetry, Differentiation, Convolution, Parseval) with proofs.
- **35:00 – 40:00 (5 mins):** Time-Frequency Duality, the Uncertainty Principle, and Check-points.

2 Mathematical Definition of the DTFT

The Discrete-Time Fourier Transform (DTFT) maps a discrete-time sequence $x[n]$ to a continuous, periodic complex frequency function $X(e^{j\omega})$:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad (1)$$

Periodicity Proof:

$$X(e^{j(\omega+2\pi)}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j(\omega+2\pi)n} = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}e^{-j2\pi n} = X(e^{j\omega}) \quad (2)$$

Since $e^{-j2\pi n} = 1$ for any integer n , the transform is periodic with period 2π .

The Inverse DTFT (IDTFT) is defined as:

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega \quad (3)$$

2.1 Convergence Conditions

A sequence $x[n]$ has a uniformly convergent DTFT if it is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty \quad (4)$$

3 DTFT of Common Signals

3.1 Finite Rectangular Pulse (Length M)

Let $x[n] = 1$ for $0 \leq n \leq M-1$, and 0 otherwise.

$$X(e^{j\omega}) = \sum_{n=0}^{M-1} e^{-j\omega n} = \frac{1 - e^{-j\omega M}}{1 - e^{-j\omega}} = e^{-j\omega(M-1)/2} \frac{\sin(\frac{\omega M}{2})}{\sin(\frac{\omega}{2})} \quad (5)$$

This is the periodic sinc function (Dirichlet kernel).

3.2 Single-Sided Decaying Exponential

Let $x[n] = a^n u[n]$ ($|a| < 1$).

$$X(e^{j\omega}) = \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}} \quad (6)$$

Magnitude:

$$|X(e^{j\omega})| = \frac{1}{\sqrt{1 + a^2 - 2a \cos \omega}} \quad (7)$$

3.3 Unit Impulse and Constant DC

For $x[n] = \delta[n]$, $X(e^{j\omega}) = 1$. For $x[n] = 1$, $X(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$.

4 Key Properties of the DTFT

- **Linearity:** $ax[n] + by[n] \leftrightarrow aX(e^{j\omega}) + bY(e^{j\omega})$
- **Time Shifting:** $x[n - n_0] \leftrightarrow X(e^{j\omega})e^{-j\omega n_0}$
- **Frequency Shifting (Modulation):** $x[n]e^{j\omega_0 n} \leftrightarrow X(e^{j(\omega - \omega_0)})$
- **Differentiation in Frequency:** $nx[n] \leftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$
- **Convolution Theorem:** $x[n] * y[n] \leftrightarrow X(e^{j\omega}) \cdot Y(e^{j\omega})$
- **Parseval's Theorem:** $\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$

5 Detailed Worked Numerical Examples

5.1 Example 1: DTFT of a Shifted Impulse Pair

Find the DTFT of $x[n] = \delta[n] - \delta[n-1]$.

$$X(e^{j\omega}) = 1 - e^{-j\omega} = e^{-j\omega/2} (e^{j\omega/2} - e^{-j\omega/2}) = 2j \sin(\omega/2) e^{-j\omega/2} = 2 \sin(\omega/2) e^{j(\pi/2 - \omega/2)}$$

Magnitude: $|X(e^{j\omega})| = 2|\sin(\omega/2)|$.

5.2 Example 2: DTFT of a Modulated Signal

Find the DTFT of $x[n] = a^n \cos(\omega_0 n) u[n]$ for $|a| < 1$. Using Euler's formula:

$$X(e^{j\omega}) = \text{DTFT} \left\{ \frac{1}{2} a^n e^{j\omega_0 n} u[n] + \frac{1}{2} a^n e^{-j\omega_0 n} u[n] \right\} = \frac{1}{2} \left[\frac{1}{1 - ae^{-j(\omega - \omega_0)}} + \frac{1}{1 - ae^{-j(\omega + \omega_0)}} \right]$$

5.3 Example 3: Energy Calculation using Parseval's

Given $x[n] = 0.5^n u[n]$, calculate total energy. In the time domain:

$$E = \sum_{n=0}^{\infty} (0.25)^n = \frac{1}{1 - 0.25} = \frac{4}{3}$$

In the frequency domain using Parseval's formula:

$$E = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1}{1.25 - \cos \omega} d\omega = \frac{1}{\sqrt{1.25^2 - 1}} = \frac{1}{\sqrt{1.5625 - 1}} = \frac{1}{\sqrt{0.5625}} = \frac{1}{0.75} = \frac{4}{3}$$

Both domains yield identical energy, verifying Parseval's theorem.